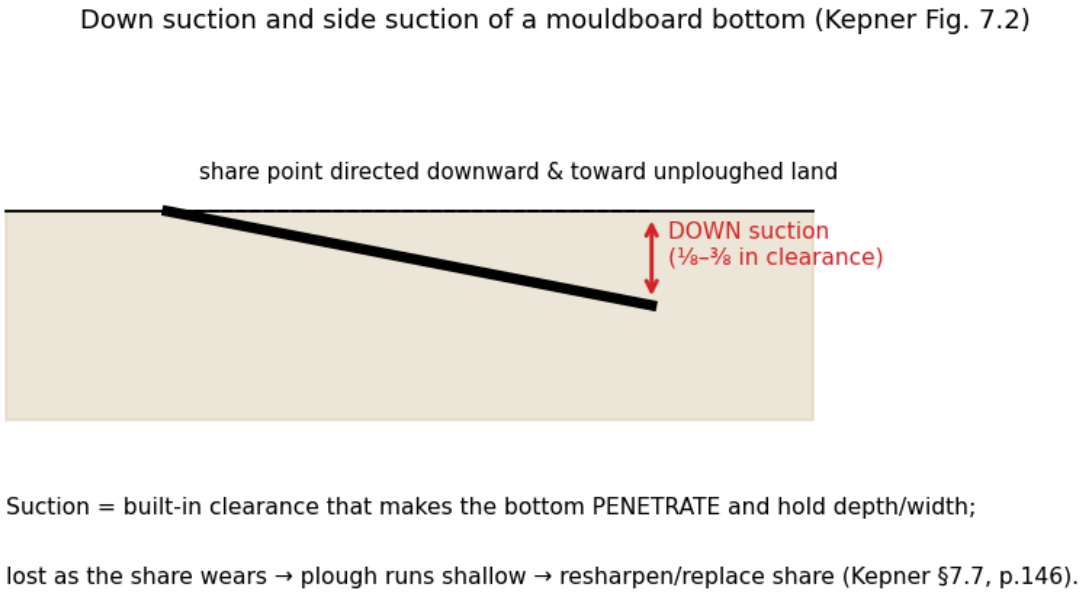
UNIT 6 · TOPIC 4

Design Parameters and Performance of Tillage Tools

Syllabus topic: Design parameters and performance of tillage tools

Mouldboard bottom — suction



Down suction of a mouldboard bottom (side suction is the matching lateral clearance)

Suction	What it is	Purpose
DOWN suction	Vertical clearance — share point dipped below the landside plane	Pulls the plough into the ground; holds DEPTH
SIDE suction	Lateral clearance — point set toward the unploughed land	Holds the WIDTH of cut

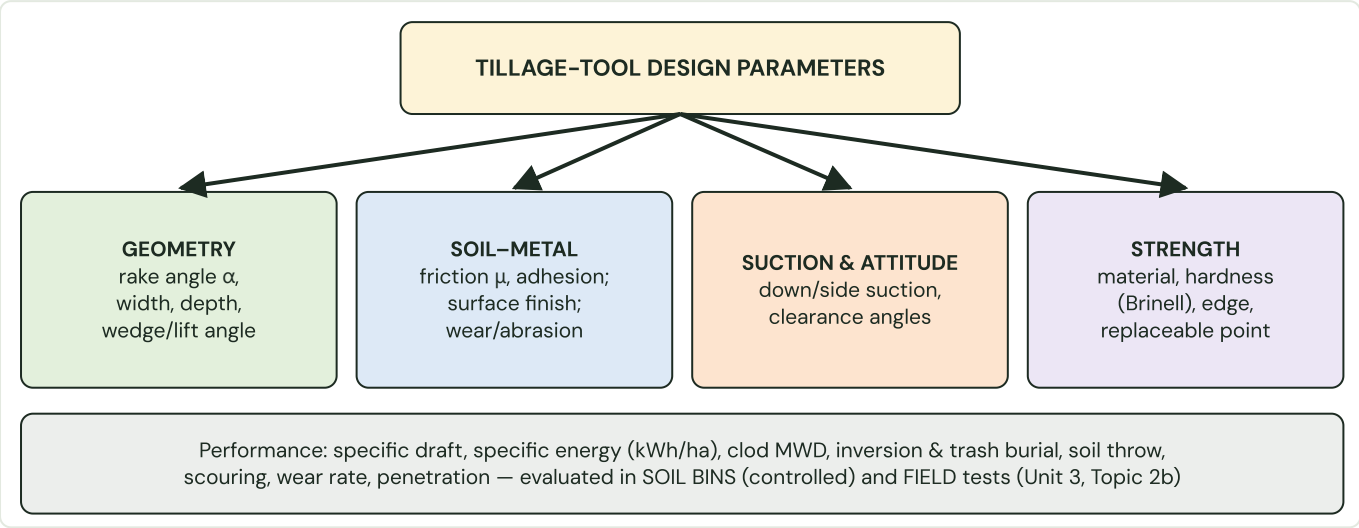
- Typical values: **⅛–⅜ inch** (≈ 3–10 mm) (Kepner p. 142).
- With too little down or side suction, the desired depth or width of cut cannot be held (p. 142).
- Suction is lost as the share wears — hence periodic sharpening or replacement.

Mouldboard bottom shapes

Bottom	Feature / use
GENERAL-PURPOSE	Moderate curvature, moderate speed, good inversion — the common Indian bottom
STUBBLE / high-speed	Short, abruptly curved — more pulverisation
SLAT	Open mouldboard — for sticky soils that will not scour
BREAKER / deep-digger	Deep work

- The mouldboard progressively cuts (share), lifts, inverts and pulverises the furrow slice; the share, shin, wing, landside and heel each have a role.
- Disc tools penetrate by WEIGHT, not suction (Unit 2, Topic 5).

Design parameters to fix



Tillage-tool design parameters and performance measures (redrawn from the source figure)

1. **Working geometry:** rake angle, width, depth, curvature/lift.
2. **Soil–metal interface:** surface finish, coatings to control friction and adhesion.
3. **Suction / attitude:** clearances and set.
4. **Structural strength:** material, hardness (Brinell), edge design, replaceable points.

Design objective: MINIMUM specific draft consistent with the required quality of work.

Performance measures

- Specific/unit draft (kN per unit furrow area or kN/m width) and specific energy (kWh/ha).
- Quality of pulverisation — clod Mean Weight Diameter (Unit 3, Topic 2b).
- Degree of soil inversion and trash burial; soil throw and ridge/furrow profile.
- Scouring behaviour; edge wear rate; penetration and depth holding.

Evaluation and prediction

- **SOIL BINS:** controlled, repeatable soil condition — vary one factor at a time and measure forces with strain-gauge (extended-octagonal-ring) dynamometers.
- **FIELD tests:** real variability (Unit 3, Topic 2b).
- **Prediction:** draft from soil strength using the models of Topic 2 — Mohr–Coulomb thrust, the ASAE D497 draft equation, the Reece equation.

Practice MCQs

Q58. Down suction of a mouldboard bottom is the:

(a) Vertical clearance dipping the share point below the landside plane (b) Lateral clearance

(c) Fuel suction (d) Air gap

Q59. Side suction helps the plough maintain:

(a) Speed
(b) Width of cut
(c) Depth only
(d) Colour

Q60. Typical down/side suction values are about:

(a) 1/8–3/8 inch
(b) 2–3 inch
(c) 6 inch
(d) Zero

Q61. Insufficient suction makes it difficult to maintain:

(a) Desired depth and width of cut
(b) Engine speed
(c) Fuel level
(d) Steering

Q62. The general-purpose mouldboard bottom is characterised by:

(a) No curvature
(b) Moderate curvature for good inversion at moderate speed
(c) Flat plate
(d) Vertical face

Q63. A slat (open) mouldboard is used for:

(a) Dry sand
(b) Sticky soils that will not scour
(c) Rock
(d) Puddling

Q64. Disc ploughs penetrate mainly by:

(a) Suction
(b) Added weight (no suction)
(c) Adhesion
(d) Vibration

Q65. Material hardness of soil-engaging parts is commonly specified by the:

(a) Rockwell C only
(b) Brinell hardness number
(c) Cone index
(d) MWD

Q66. The size-independent measure to compare tillage-tool energy use is:

(a) Total draft
(b) Specific draft / specific energy (kWh/ha)
(c) Mass
(d) Width

Q67. Quality of pulverization is quantified by:

(a) Clod Mean Weight Diameter
(b) VMD
(c) Slip
(d) Cone apex

Q68. Controlled, repeatable evaluation of tillage tools is done in a:

(a) Wind tunnel
(b) Soil bin
(c) Patternator
(d) Calorimeter

Q69. Soil-bin tool forces are measured with:

(a) Thermometers
(b) Strain-gauge (extended-octagonal-ring) dynamometers
(c) Hydrometers
(d) Anemometers

Q70. Tillage-tool draft can be predicted from soil strength using:

(a) Mohr–Coulomb / ASAE D497 / Reece models
(b) Ohm's law
(c) Bernoulli
(d) Nusselt number

Answers

Q58 — (a) Vertical clearance dipping the share point below the landside plane — *Kepner p. 142 — down suction is vertical clearance.*

Q59 — (b) Width of cut

Q60 — (a) 1/8–3/8 inch — *Kepner p. 142 — ⅛–⅜ in.*

Q61 — (a) Desired depth and width of cut — *Without suction, depth/width cannot be held.*

Q62 — (b) Moderate curvature for good inversion at moderate speed
Q63 — (b) Sticky soils that will not scour
Q64 — (b) Added weight (no suction)
Q65 — (b) Brinell hardness number
Q66 — (b) Specific draft / specific energy (kWh/ha)
Q67 — (a) Clod Mean Weight Diameter
Q68 — (b) Soil bin
Q69 — (b) Strain-gauge (extended-octagonal-ring) dynamometers
Q70 — (a) Mohr–Coulomb / ASAE D497 / Reece models

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 6–7 (force analysis; moldboard tools), pp. 116–169; Gill & Vanden Berg — USDA Agriculture Handbook 316; Nichols' soil–metal friction/shear studies (cited in Kepner); Reece universal earthmoving equation; O'Callaghan & Payne tine mechanics; NPTEL soil-dynamics lectures.